



Exam Room Item Identifier

An AI-powered app that identifies items brought into the exam room to help keep exams fair, safe, and secure.



Fair Exams.
Safe Environment.
Trusted Results.



1. WHY DO WE NEED THIS APP?

Exams should be a fair way to measure knowledge. However, some items brought into the exam room—such as phones, smartwatches, or unauthorized devices—can give unfair advantages or pose safety risks. Our app uses AI to quickly and accurately identify items, helping invigilators maintain a secure and fair exam environment.



Promotes academic integrity



Prevents cheating and rule violations



Helps detect dangerous or prohibited items



Saves time and reduces manual checks

2. PROBLEMS WE FACED



1. We don't have enough dataset.

We initially had very few images, which made it hard for the AI to learn and recognize many items.



2. Hard to identify which is calculator and which is phone.

Calculators and phones look similar in shape and size, so the AI often got confused.



3. Can't recognise dangerous things.

The AI struggled to detect dangerous or prohibited items because we had very few or no examples of those items in our dataset.

3. HOW WE BUILT THE APP



Find Dataset

We collected images from online sources, created our own data by taking photos, and used data augmentation to increase the number and variety of images.



Train AI Model

We used a machine learning model (YOLO) to learn from the images. The model was trained repeatedly to improve its accuracy.



Fix Errors and Improve

We analyzed the mistakes, added more data for weak classes, adjusted the model, and tested again to make it better.

4. PROBLEMS WE HAVE SOLVED (SO FAR)



1. We have more dataset now.

We collected and created a much larger and more diverse dataset, which helped the AI learn better and recognize more items.



2. Better at telling calculator and phone apart.

By adding many examples and fine-tuning the model, the AI can now accurately tell the difference between calculators and phones.



3. Improved at recognizing dangerous items.

We added images of dangerous and prohibited items (e.g., knives, other electronic devices), so the AI can now detect more of these items.



OUR GOAL

To keep improving the app so it can identify more items with higher accuracy and help make every exam fair and safe for everyone.

WHAT'S NEXT?



Collect even more data and add more item types



Improve accuracy in different lighting and angles



Make the app faster and easier to use



Enable real-time alerts and cloud reporting



Smart Detection.
Fair Protection.
Better Education.

